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Turbomachine operation monitoring system and method

Abstract

A method of monitoring the operation of a turbomachine includes positioning a flux probe in an operating position with respect to a rotor of the turbomachine, rotating the rotor at an operating speed with respect to a stator to one of generate an electrical output and produce an output torque. The method also includes measuring the passage of each slot of a plurality of slots formed in the rotor using the flux probe, calculating a time duration between the passage of a first slot of the plurality of slots and a second slot of the plurality of slots, and comparing an expected time duration for the operating speed with the calculated time duration. A difference in the comparison is indicative of one of a power change and a torsional vibration condition.

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Background/Summary

BACKGROUND

(1) Turbomachines include motors and generators. Generators are often used for power generation activities to generate grid-suitable electricity using a prime mover such as a gas turbine, steam turbine, wind turbine, hydro turbine, and the like to drive the generator. The generators generally include a stator that remains stationary during operation and a rotor that rotates with respect to the stator. The rotor often includes two or more poles that when rotated interact with the stator to generate the desired current at the desired frequency and voltage.

BRIEF SUMMARY

(2) In one aspect, a method of monitoring the operation of a turbomachine includes positioning a flux probe in an operating position with respect to a rotor of the turbomachine, rotating the rotor at an operating speed with respect to a stator to one of generate an electrical output and produce an output torque. The method also includes measuring the passage of each slot of a plurality of slots formed in the rotor using the flux probe, calculating a time duration between the passage of a first slot of the plurality of slots and a second slot of the plurality of slots, and comparing an expected time duration for the operating speed with the calculated time duration. A difference in the comparison is indicative of one of a power change and a torsional vibration condition.

(3) In one aspect, a method of determining a power level for a turbomachine includes positioning a

flux probe in an operating position with respect to a rotor of the turbomachine, operating the turbomachine to rotate the rotor at an operating speed with respect to a stator, and measuring a flux using the flux probe for a full rotation of the rotor. The method also includes calculating a flux density curve for each rotation of the rotor, determining a first position of the rotor with respect to the flux probe when the value of the flux density curve equals zero, and determining the power level based solely on the first position.

(4) In one aspect, a turbomachine system includes a flux probe in an operating position with respect to a rotor of the turbomachine, the rotor rotatable at an operating speed with respect to a stator to one of generate an electrical output and produce an output torque. The turbomachine system also includes a processor and a memory storing instructions that, when executed by the processor, configure the turbomachine system to measure the passage of each slot of a plurality of slots formed in the rotor using the flux probe, calculate a time duration between the passage of a first slot of the plurality of slots and a second slot of the plurality of slots, and compare an expected time duration for the operating speed with the calculated time duration. A difference in the comparison is indicative of one of a power change and a torsional vibration condition.

(5) The foregoing has broadly outlined some of the technical features of the present disclosure so that those skilled in the art may better understand the detailed description that follows. Additional features and advantages of the disclosure will be described hereinafter that form the subject of the claims. Those skilled in the art will appreciate that they may readily use the conception and the specific embodiments disclosed as a basis for modifying or designing other structures for carrying out the same purposes of the present disclosure. Those skilled in the art will also realize that such equivalent constructions do not depart from the spirit and scope of the disclosure in its broadest form.

(6) Also, before undertaking the Detailed Description below, it should be understood that various definitions for certain words and phrases are provided throughout this patent document, and those of ordinary skill in the art will understand that such definitions apply in many, if not most, instances to prior as well as future uses of such defined words and phrases. While some terms may include a wide variety of embodiments, the appended claims may expressly limit these terms to specific embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced.

(2) FIG. 1 is a cross-sectional view of a generator taken along the generator centerline, rotational, or longitudinal axis.

(3) FIG. 2 is a perspective view of a rotor suitable for use in the generator of FIG. 1.

(4) FIG. 3 is a perspective view of a portion of a stator suitable for use in the generator of FIG. 1.

(5) FIG. 4 is a schematic illustration of a rotor arrangement including a flux probe in an operating position.

(6) FIG. 5 is a chart illustrating the data collected by the flux probe of FIG. 4 for one rotation of the rotor at a no load or zero power output condition.

(7) FIG. 6 is a chart illustrating the data collected by the flux probe of FIG. 4 for one rotation of the rotor at a medium load or medium power output condition

(8) FIG. 7 is a chart illustrating the data collected by the flux probe of FIG. 4 for one rotation of the rotor at a high load or high power output condition.

DETAILED DESCRIPTION

(9) Before any embodiments of the invention are explained in detail, it is to be understood that the

invention is not limited in its application to the details of construction and the arrangement of components set forth in this description or illustrated in the following drawings. The invention is capable of other embodiments and of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting.

(10) Various technologies that pertain to systems and methods will now be described with reference to the drawings, where like reference numerals represent like elements throughout. The drawings discussed below, and the various embodiments used to describe the principles of the present disclosure in this patent document are by way of illustration only and should not be construed in any way to limit the scope of the disclosure. Those skilled in the art will understand that the principles of the present disclosure may be implemented in any suitably arranged apparatus. It is to be understood that functionality that is described as being carried out by certain system elements may be performed by multiple elements. Similarly, for instance, an element may be configured to perform functionality that is described as being carried out by multiple elements. The numerous innovative teachings of the present application will be described with reference to exemplary non-limiting embodiments.

(11) It should be understood that the words or phrases used herein should be construed broadly, unless expressly limited in some examples. For example, the terms “including,” “having,” and “comprising,” as well as derivatives thereof, mean inclusion without limitation. The singular forms “a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Further, the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. The term “or” is inclusive, meaning and/or, unless the context clearly indicates otherwise. The phrases “associated with” and “associated therewith,” as well as derivatives thereof, may mean to include, be included within, interconnect with, contain, be contained within, connect to or with, couple to or with, be communicable with, cooperate with, interleave, juxtapose, be proximate to, be bound to or with, have, have a property of, or the like. Furthermore, while multiple embodiments or constructions may be described herein, any features, methods, steps, components, etc. described with regard to one embodiment are equally applicable to other embodiments absent a specific statement to the contrary.

(12) Also, although the terms “first,” “second,” “third” and so forth may be used herein to refer to various elements, information, functions, or acts, these elements, information, functions, or acts should not be limited by these terms. Rather these numeral adjectives are used to distinguish different elements, information, functions or acts from each other. For example, a first element, information, function, or act could be termed a second element, information, function, or act, and, similarly, a second element, information, function, or act could be termed a first element, information, function, or act, without departing from the scope of the present disclosure.

(13) In addition, the term “adjacent to” may mean that an element is relatively near to but not in contact with a further element or that the element is in contact with the further portion, unless the context clearly indicates otherwise. Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise. Terms “about” or “substantially” or like terms are intended to cover variations in a value that are within normal industry manufacturing tolerances for that dimension. If no industry standard is available, a variation of twenty percent would fall within the meaning of these terms unless otherwise stated.

(14) As illustrated in FIG. 1, a generator **100** includes a stator **300** and a rotor **200** supported for rotation within the stator **300**. The stator **300** includes a stator housing **102** that surrounds and substantially encloses a stator core **104**. The stator core **104** is often made-up of a number of laminations **106** stacked in a longitudinal direction (along a rotational axis). Each lamination **106** includes cut outs or is otherwise shaped to define the desired features of the rotor core, including a bore **108** that is sized to receive the rotor **200**.

(15) In some constructions, a stator cooling system **110** is provided to cool the stator **300** and improve the efficiency and power density of the stator **300**. In some constructions, a cooling gas is employed as a stator coolant. However, larger stators **300** may include liquid cooling such as water cooling.

(16) The rotor **200** includes a rotor core **112**, a rotor shaft **114**, and two retaining rings **116** coupled to the rotor shaft **114**. The illustrated rotor shaft **114** is supported for rotation by a bearing **118** positioned at each end of the rotor **200**. A turbine coupling **120** is positioned at one end of the rotor **200** to facilitate connection of the rotor **200** to a turbine (e.g., combustion turbine, steam turbine, hydro turbine, wind turbine, etc.) or to another prime mover. The opposite end of the rotor **200** may include an exciter coupling **122** that allows for connection to an exciter or other rotating equipment.

(17) The generator **100** illustrated in FIG. **1** is a synchronous generator **100**. However, asynchronous generators or motors could include the features described herein.

(18) FIG. **2** illustrates the rotor **200** of FIG. **1** in greater detail. The rotor core **112** includes a series of rotor slots **202** that extend longitudinally along the rotor core **112**. Rotor windings **204** are positioned within the rotor slots **202** to define one or more pairs of poles. In the illustrated construction two poles are formed by the rotor windings **204**. However, other constructions could include four poles, eight poles or more poles if desired. The rotor **200**, sometimes referred to as a field, also includes a commutator **206** that provides a connection to an exciter that provides electrical current at a desired voltage to the windings **302** to generate a magnetic field.

(19) The rotor **200** may also include a rotor cooling system **208** that operates to cool the rotor **200**. In some constructions, the rotor **200** is air-cooled with other constructions employing another fluid such as hydrogen.

(20) Turning to FIG. **3**, the stator core **104** is illustrated in greater detail. The stator core **104**, in most constructions is formed from a series of laminations **106** that are stacked in the longitudinal direction. Each lamination **106** includes cut-outs, openings or features that when stacked define the desired features of the stator core **104**. Each of the laminations **106** includes a number of teeth **304** that are evenly spaced circumferentially around the bore **108**. the teeth **304** cooperate to define a series of slots that extend the length of the stator core **104**. Bars **306** are positioned within the slots and are electrically connected to one another to define a series of windings **302**. In the illustrated generator **100**, the windings **302** are arranged to define three phases. Generally, the three phases are electrically arranged to define a delta-circuit, or a Y-circuit as may be desired. Of course, other constructions could include a single phase if desired.

(21) As part of the stator cooling system **110**, each of the bars **306** includes one or more coolant passages **308** that allow for the flow of coolant along the length of the bar **306**. As discussed, a coolant fluid such as water is often employed to cool the generator.

(22) Before discussing FIG. **4** in detail, it should be noted that the system described herein is applicable to virtually any electrical turbomachine. Specifically, it is applicable to virtually any type of generator and motor. When the turbomachine operates as a motor, a current is applied to the rotor **200** and power is directed to the stator **300** to generate a usable torque at the rotor **200**. When operated as a generator, a current and a torque are applied to the rotor **200** to generate a usable electrical power output at the stator **300**. The remainder of this description will focus on an example that is operated as a generator. However, it should be clear that it is equally applicable to a motor.

(23) FIG. **4** illustrates a rotor arrangement **400** suitable for use in a system that monitors certain operating parameters of the rotor **200** using a flux probe **402**. The rotor **200** includes a rotor body **404** that is generally formed from a solid or laminated ferromagnetic material. A plurality of slots **406** are formed in the body **404** and extend along a portion of the length of the body **404**. In the rotor **200** of FIG. **4**, the slots **406** are arranged to define two poles, an A-pole **408** and a B-pole **410**. Other constructions may include slots **406** arranged to define four poles, or more than four poles as may be desired.

(24) The slots **406** are arranged in pairs with a leading slot formed on one side of a pole and a lagging slot formed on the opposite side of the pole. The leading and lagging sides of the pole are defined with respect to the direction of rotation **412** of the rotor **200**. The leading side passes a fixed point (e.g., the flux probe **402**) on any rotation before the lagging side. With reference to the rotor **200** of FIG. 4, leading A-slots **414** are formed on the leading side of the A-pole **408** with a corresponding number of lagging A-slots **416** formed on the opposite side of the A-pole **408**. Similarly, leading B-slots **418** are formed on the leading side of the B-pole **410** with lagging B-slots **420** formed on the opposite side of the B-pole **410**. In the illustrated construction, six slots **406** are employed on each side of each pole with other constructions having more or fewer slots **406** as may be desired.

(25) To form the electrical circuit in the rotor **200**, windings **302** are placed in each of the slots **406**. Each winding **302** is a complete loop that extends around on of the A-pole **408** or the B-pole **410**. As such, each winding **302** includes a leading portion disposed in a leading slot **406** and a lagging portion disposed in a lagging slot **406**. In addition, multiple windings **302** are generally stacked on top of one another within each of the slots **406** to provide the number of turns desired around each of the poles. With reference to the example of FIG. 4, leading A-windings **422** are positioned in the leading A-slots **414** and lagging A-windings **424** are placed in the lagging A-slots **416**. Similarly, leading B-windings **426** are positioned in the leading B-slots **418** and lagging B-windings **428** are placed in the lagging B-slots **420**.

(26) During operation, a current **430** is passed through each of the windings **302**. The current **430** flows through each of the windings **302** in a loop around its respective pole to magnetize each of the A-pole **408** and B-pole **410** thereby producing a rotating magnetic field when the rotor **200** is rotated. As the current **430** flows along the length of the slots **406** it produces a magnetic flux **432** at the top or outermost surface of each slot **406**. The direction of the flux **432** at each slot **406** is a function of the direction of the current **430** in the windings **302** disposed in the slots **406**. In the example of FIG. 4, the current **430** is flowing in a direction out of the page on the left side and into the page on the right side. This leads to counterclockwise fluxes **432** on the left side of the rotor **200** and clockwise fluxes **432** on the right side of the rotor **200**.

(27) The flux probe **402** is positioned adjacent the rotor **200** and in close enough proximity to detect these fluxes **432** as each of the slots **406** rotates past the flux probe **402**. The actual circumferential position of the flux probe **402** with respect to the rotor is not relevant so long as it is capable of measuring a full rotation of the rotor **200** and that position is known to a computer system **434** that will ultimately analyze the data generated by the flux probe. The computer system **434** could be part of an engine control system or part of a separate system used only to monitor and analyze the outputs from the flux probe **402** as may be desired.

(28) For FIG. 5 through FIG. 7, counterclockwise fluxes **432** will be deemed positive and clockwise fluxes **432** will be deemed negative. Of course, this could be reversed if desired. As such, the terms “positive” and “negative” refer to the example provided but should not be limiting as they could be reversed simply by changing the positive and negative values for the flux **432**.

(29) FIG. 5 includes a chart **500** showing the data collected by the flux probe **402** for one rotation plotted as a flux curve **502**. The flux curve **502** is plotted with time (in microseconds) along the X-axis and a normalized amplitude of the flux along the Y-axis. The illustrated flux curve **502** includes one full rotation of the rotor (i.e., about 0.017 seconds at 3600 RPM, or 0.02 seconds at 3000 RPM). In addition, a flux density curve **504** is plotted for the same time frame. The flux density curve **504** is calculated from the data collected to develop the flux curve **502**. In preferred constructions, a sampling rate of 0.1 ms is employed for the flux probe **402** with faster or slower sampling rates being possible.

(30) The flux curve **502** illustrates data collected from the B-pole **410** between time “zero” and 6 ms. Specifically, this portion of the flux curve **502** illustrates the measured flux from a lagging B-pole **506**. The central portion of the flux curve **502** (between about 6 ms and 14.5 ms) illustrates

data collected from the A-pole **408**. The remainder of the flux curve **502** illustrates the other portion of the B-pole **410** and specifically the data from a leading B-pole **508**.

(31) The flux probe **402** measures peaks or spikes (positive or negative) of flux as each slot **406** passes the flux probe **402**. The largest of these spikes is typically at a slot **406** that only includes one immediately adjacent slot **406**. Specifically, with reference to the positive portion of the flux curve **502** there is a large spike as a first lagging B-slot **436** passes the flux probe **402** and as a first leading A-slot **438** passes. A measured time period **510** between the passage of the first lagging B-slot **436** and the first leading A-slot **438** can be compared to a calculated time period for the rotor **200** to rotate the same distance to make some evaluations regarding the status of the rotor **200**. For example, for a rotor **200** with the first lagging B-slot **436** and the first leading A-slot **438** separated by about 115 degrees and that should rotate at 3600 rpm, a calculated time period may be about 5.32 ms. If the comparison of the calculated time period and the measured time period **510** shows that the measured time period **510** is longer than the expected or calculated time period, one can conclude that the rotor **200** is rotating slightly slower than expected. This may be indicative of a load increase or a torsional vibration. If the difference slowly decays to zero with subsequent revolutions, it might be indicative of a load change. If the difference varies with each revolution, it may be indicative of a torsional vibration. In the case of a torsional vibration, repeated measurements from subsequent rotations can be used to estimate a vibration frequency. In addition to this comparison, the measured time period **510** (sometimes referred to as calculated time period) can be used to calculate an actual rotating speed of the rotor **200** which can be provided to a user.

(32) Additional information can be extracted from the chart **500** of FIG. 5. The flux density curve **504** moves in a generally sinusoidal pattern, oscillating from a positive peak to a negative peak and back to a positive peak with each rotation. Because of this oscillation, the flux density curve **504** crosses a zero amplitude twice during each rotation. This point is illustrated as a pair of zero flux density times **512** illustrated on the chart **500**. As will be better illustrated in the upcoming figures, the position of the rotor **200** when this crossing occurs is indicative of the load or the power output of the generator **100**.

(33) FIG. 5 illustrates a generator **100** operating in a no load or zero power output condition. In this operating condition, the zero flux density times **512** occur directly between the passage of the A-pole **408** and the B-pole. In other words, the rotor **200** has rotated one of the poles, in FIG. 5 the B-pole **410** ninety degrees past the flux probe **402**.

(34) FIG. 6 illustrates a second chart **600** of the data collected by the flux probe **402** for a rotation of the rotor **200** operating at a medium power output or load. As can be seen, the shape of the flux curve **502** has changed slightly which changes the shape of the flux density curve **504** and shifts the zero flux density times **512** to the right or toward the pole in front of the zero flux density time **512**.

(35) FIG. 7 illustrates a third chart **700** of the data collected by the flux probe **402** for a rotation of the rotor **200** operating at a high power output or load. As can be seen, the shape of the flux curve **502** has changed slightly which changes the shape of the flux density curve **504** and shifts the zero flux density times **512** further to the right or toward the pole in front of the zero flux density time **512**.

(36) It should be noted that the measured time period **510** between the first lagging B-slot **436** and the first leading A-slot **438** does not change with the changing load such that this measured time period **510** can be a reliable indication of speed.

(37) In operation, an exciter or other system provides current at a desired voltage to the rotor **200**. The current flows through the rotor windings **204** to establish two magnetic poles in a two-pole generator and more poles in higher pole generators. The turbine, or other prime mover is coupled to the rotor **200** and operates to rotate the rotor **200** at a desired speed. For a synchronous generator with a two-pole rotor **200**, the rotor is rotated at 3600 RPM to generate 60 Hz electricity. For electricity at 50 Hz, the rotor **200** is rotated at 3000 RPM.

(38) Although an exemplary embodiment of the present disclosure has been described in detail,

those skilled in the art will understand that various changes, substitutions, variations, and improvements disclosed herein may be made without departing from the spirit and scope of the disclosure in its broadest form.

(39) None of the description in the present application should be read as implying that any particular element, step, act, or function is an essential element, which must be included in the claim scope: the scope of patented subject matter is defined only by the allowed claims. Moreover, none of these claims are intended to invoke a means plus function claim construction unless the exact words “means for” are followed by a participle.

Claims

1. A method of monitoring the operation of a turbomachine, the method comprising: positioning a flux probe in an operating position with respect to a rotor of the turbomachine; rotating the rotor at an operating speed with respect to a stator to one of generate an electrical output and produce an output torque; measuring the passage of each slot of a plurality of slots formed in the rotor using the flux probe; calculating a time duration between the passage of a first slot of the plurality of slots and a second slot of the plurality of slots; comparing an expected time duration between the passage of the first slot and the second slot when operating at an operating speed with the calculated time duration, a difference in the comparison indicative of one of a power change and a torsional vibration condition, wherein the rotor defines an A-pole having a first A-winding disposed in a first A-slot, the first A-slot including a first leading A-slot and a first lagging A-slot, and a B-pole having a first B-winding disposed in a first B-slot, the first B-slot including a first leading B-slot and a first lagging B-slot, and wherein the first slot is the first lagging B-slot and the second slot is the first leading A-slot.

2. The method of claim 1, wherein the rotor defines an A-pole having a plurality of A-windings disposed in a corresponding plurality of A-slots, and a B-pole having a plurality of B-windings disposed in a corresponding plurality of B-slots, the method further comprising calculating a flux density curve using the flux probe, the flux density curve having an amplitude of zero at least twice per rotation of the rotor.

3. The method of claim 2, further comprising determining the position of the plurality of A-windings and the plurality of B-windings at the time the amplitude of the flux density curve is zero, the position of the plurality of A-windings and the plurality of B-windings being indicative of a power output of the turbomachine.

4. The method of claim 3, wherein the plurality of A-windings are divided into a plurality of leading A-windings and a plurality of trailing A-windings, and wherein the A-pole is disposed between the plurality of leading A-windings and the plurality of trailing A-windings, and wherein the time between the A-pole passing the flux probe and the next time the amplitude of the flux density curve is zero decreases as power increases.

5. The method of claim 2, further comprising determining that the turbomachine is operating in a no-load condition in response to the flux probe detecting the passage of one of the A-pole and the B-pole one-quarter of a rotational period after the time the amplitude of the flux density curve is zero.

6. A method of determining a power level for a turbomachine, the method comprising: positioning a flux probe in an operating position with respect to a rotor of the turbomachine; operating the turbomachine to rotate the rotor at an operating speed with respect to a stator; measuring a flux using the flux probe for a full rotation of the rotor; calculating a flux density curve for each rotation of the rotor; determining a first position of the rotor with respect to the flux probe when the value of the flux density curve equals zero; and determining the power level based solely on the first position, wherein the rotor defines an A-pole having a first A-winding disposed in a first A-slot, the first A-slot including a first leading A-slot and a first lagging A-slot, and a B-pole having a first B-

winding disposed in a first B-slot, the first B-slot including a first leading B-slot and a first lagging B-slot, the method further comprising determining a time duration between the time the first lagging B-slot passes the flux probe and the time the first leading A-slot passes the flux probe and comparing that time to an expected time duration, a difference between the determined time duration and the expected time duration indicative of one of a power change and a torsional vibration condition.

7. The method of claim 6, wherein the rotor defines an A-pole having a plurality of A-windings disposed in a corresponding plurality of A-slots, and a B-pole having a plurality of B-windings disposed in a corresponding plurality of B-slots, the flux density curve having an amplitude of zero at least twice per rotation of the rotor.

8. The method of claim 7, further comprising determining the position of the plurality of A-windings and the plurality of B-windings at the time the amplitude of the flux density curve is zero, the position of the plurality of A-windings and the plurality of B-windings at the time the amplitude of the flux density curve is zero being indicative of the power level of the turbomachine.

9. The method of claim 8, wherein the plurality of A-windings are divided into a plurality of leading A-windings and a plurality of trailing A-windings, and wherein the A-pole is disposed between the plurality of leading A-windings and the plurality of trailing A-windings, and wherein the time between the A-pole passing the flux probe and the next time the amplitude of the flux density curve is zero decreases as power increases.

10. The method of claim 7, further comprising determining that the turbomachine is operating in a no-load condition in response to the flux probe detecting the passage of one of the A-pole and the B-pole one-quarter of a rotational period after the time the amplitude of the flux density curve is zero.

11. A turbomachine system comprising: a flux probe in an operating position with respect to a rotor of the turbomachine, the rotor rotatable at an operating speed with respect to a stator to one of generate an electrical output and produce an output torque; a processor; and a memory storing instructions that, when executed by the processor, configure the turbomachine system to: measure the passage of each slot of a plurality of slots formed in the rotor using the flux probe; calculate a time duration between the passage of a first slot of the plurality of slots and a second slot of the plurality of slots; and compare an expected time duration for the passage of the first slot and the second slot when operating at an operating speed with the calculated time duration, a difference in the comparison indicative of one of a power change and a torsional vibration condition, wherein the rotor defines an A-pole having a first A-winding disposed in a first A-slot, the first A-slot including a first leading A-slot and a first lagging A-slot, and a B-pole having a first B-winding disposed in a first B-slot, the first B-slot including a first leading B-slot and a first lagging B-slot, and wherein the first slot is the first lagging B-slot and the second slot is the first leading A-slot.

12. The turbomachine system of claim 11, wherein the rotor defines an A-pole having a plurality of A-windings disposed in a corresponding plurality of A-slots, and a B-pole having a plurality of B-windings disposed in a corresponding plurality of B-slots, wherein the instructions further configure the turbomachine system to calculate a flux density curve using the flux probe, the flux density curve having an amplitude of zero at least twice per rotation of the rotor.

13. The turbomachine system of claim 12, wherein the instructions further configure the turbomachine system to determine the position of the plurality of A-windings and the plurality of B-windings at the time the amplitude of the flux density curve is zero, the position of the plurality of A-windings and the plurality of B-windings at the time the amplitude of the flux density curve is zero being indicative of a power output of the turbomachine.

14. The turbomachine system of claim 13, wherein the plurality of A-windings are divided into a plurality of leading A-windings and a plurality of trailing A-windings, and wherein the A-pole is disposed between the plurality of leading A-windings and the plurality of trailing A-windings, and wherein the time between the A-pole passing the flux probe and the next time the amplitude of the

flux density curve is zero decreases as power increases.

15. The turbomachine system of claim 12, wherein the instructions further configure the turbomachine system to determine that the turbomachine is operating in a no-load condition in response to the flux probe detecting the passage of one of the A-pole and the B-pole one-quarter of a rotational period after the time the amplitude of the flux density curve is zero.
